OIT-367 Regression Project

Predicting Commercial Success of Songs

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Executive Summary

I. Problem

What makes a song a hit? Music label A&R (artists and repertoire) teams have grappled with this question for decades, but it is rarely examined through a rigorous, data-driven lens. This project explores what predicts commercial music success using Spotify audio features augmented with artist-level signals from the Spotify API and Last.fm. From a dataset of over 78,000 unique tracks (of which roughly 3% appeared on the Billboard Hot 100) we address two questions: what predicts whether a track reaches the chart, and among those that do, what predicts how long it stays.

II. Principal Findings

Chart Entry: Artist popularity, not audio production, is what gets a song onto the Billboard Hot 100. An artist's historical peak popularity and listener base dominate every audio feature combined. Models using only audio characteristics perform barely better than random given how rare chart entry is. Once we incorporate artist-level features, our best model (XGBoost Decision Trees) recovers a large share of genuine chart hits while minimizing false positives compared to a random baseline, a practical screening tool for music labels to allocate resources that dramatically narrows the funnel. Among audio signals, instrumentalness is the strongest barrier to charting, while higher valence, US artist origin, and explicit content provide modest independent advantages.

Chart Longevity: Genre and artist identity explain far more about how long a track stays on the Hot 100 than its audio features. Audio features alone account for roughly 5% of the variation in chart duration; adding genre and artist controls raises that to about 25%. A Cox survival model confirms that the streaming era has fundamentally compressed chart lifespans: tracks from more recent decades rotate off the chart significantly faster on a per-week basis than their predecessors. Louder tracks and those from artists with higher historical popularity tend to last longer, but even our richest models fail when asked to predict forward in time. Every model we tested performed worse than simply guessing the average when applied to songs released after 2015, indicating that what predicted chart staying power in the past does not carry over to future releases.

III. Recommendations

Record labels should invest in understanding an artist's existing audience reach before evaluating their sound. Audio production matters at the margin, particularly avoiding overly instrumental or low-valence tracks, but artist platform presence is the primary driver of chart success. For chart entry screening, our model offers a tool to narrow the funnel of tracks worth closer evaluation, though given limitations about the efficacy of historical data, it should be used with caution, rather than replace A&R judgment.

For chart longevity, labels should not rely on predictive models to forecast how long a new release will chart. The music industry's rapid structural evolution through streaming platforms, playlist algorithms, and social media dynamics makes historical patterns unreliable guides to future performance. Instead, labels should use our descriptive findings directionally: expect shorter chart windows in the streaming era, and recognize that sustained chart presence is increasingly a function of promotion infrastructure and audience scale rather than song-level audio characteristics.

I. Introduction

The Billboard Hot 100 is the primary commercial benchmark of a song's commercial success in the United States, which combines radio, download, and streaming signals into a single weekly chart. Understanding the determinants of chart entry and longevity has significant commercial implications: record labels, streaming platforms, and artists spend billions of dollars annually on music production and promotion with limited ability to predict which attributes drive commercial success.

Recent developments in music streaming have created an unprecedented opportunity for data-driven analysis. Spotify's audio feature API provides quantifiable acoustic descriptors, such as valence, instrumentalness, danceability, tempo, and others, for over 100 million tracks. These features, combined with artist popularity metrics, offer a rich basis for modeling song chart success.

This project addresses two specific research questions:

Chart Entry: *Which audio features and song attributes predict whether a track reaches the Hot 100?*

Chart Longevity: *Among tracks that chart, which features predict how long they remain on the Hot 100?*

II. Data

2.1 Sources and Feature Selection

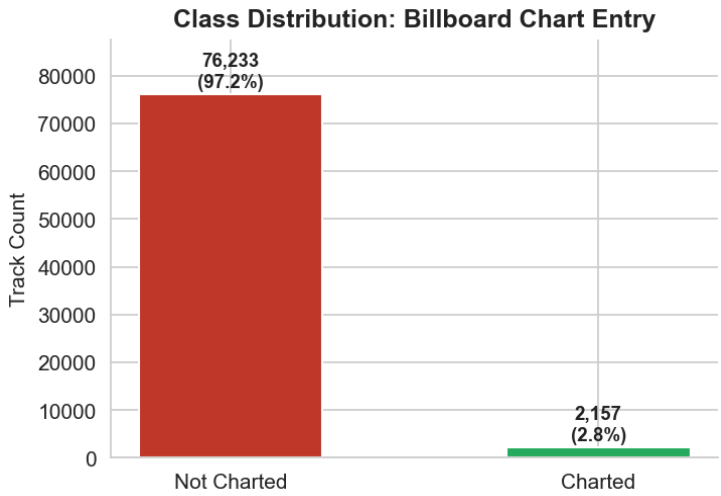
We created the base dataset by integrating the Spotify Tracks Dataset with Billboard Hot 100 snapshots (1958-2024), joined via unique Spotify track IDs and further de-depuplicated via cross-ID fuzzy matching which eliminated 1,345 duplicate charted tracks (e.g., identical songs listed under different Spotify IDs like a single vs. an album version). We augmented this dataset with artist-level signals from the Spotify Web API (historical popularity and catalog size) and log-transformed listener counts from MusicBrainz/Last.fm to account for heavy right skew.

For a subset of 856 out of 2,157 charted tracks, we incorporated VADER sentiment scores derived from song lyrics. To ensure model stability, we removed energy, danceability, and tempo after observing Correlation Matrix patterns (see Appendix for final matrix) and Variance Inflation Factors (VIF) exceeding 10. The final classification set comprises 15 features: 10 Spotify audio metrics and 5 artist-level commercial signals. Longevity models further include a decade index and lyric sentiment features (Full VIF details and feature definitions are provided in the Appendix).

2.2 Dataset Construction and Handling Imbalance

We used a data cleaning pipeline to deduplicate tracks to a single row per ID, resulting in a total of 78,390 unique songs. A left join with Billboard history identified 2,157 charted tracks (2.75% positive rate). This inherent class imbalance (illustrated in Figure 1) makes Precision-Recall AUC (PR-AUC) our primary evaluation metric. While ROC-AUC can be misleadingly optimistic by over-weighting the model's ability to correctly identify the vast majority of non-charted tracks, PR-AUC focuses specifically on the "charted" minority class. It provides a more rigorous measure of the model's ability to minimize false positives in an environment where a random classifier achieves a baseline of only 0.0275. See Appendix for full data cleaning pipeline.

Figure 1: Class Distribution (Charted vs. Not Charted Tracks)



III. Methods

3.1 Classification: Chart Entry (RQ1)

To model chart entry, we utilized a stratified 80/20 train-test split. We first established a Logistic Regression baseline to derive standardized coefficients and odds ratios for stakeholder interpretability. To capture nonlinearities and feature interactions, we implemented XGBoost 2.1.3, utilizing the SHAP (SHapley Additive exPlanations) framework to rank feature importance (see Figure 2 for final feature importance distribution). We addressed class imbalance via weighted loss functions in both models. We also trained audio-only baselines to isolate the marginal predictive lift provided by artist enrichment features. Addition of artist features was critical for achieving strong performance in classification (see Figure 3 for comparative performance over baseline).

Figure 2: SHAP Feature Importance Distribution

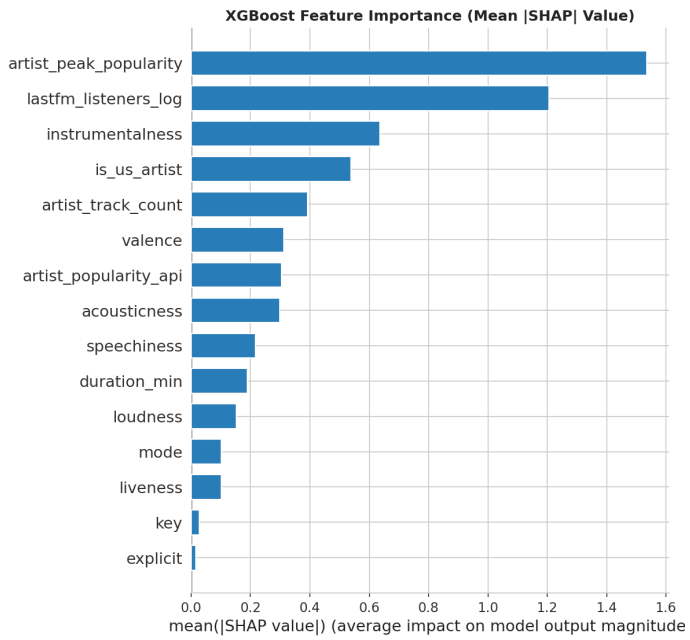
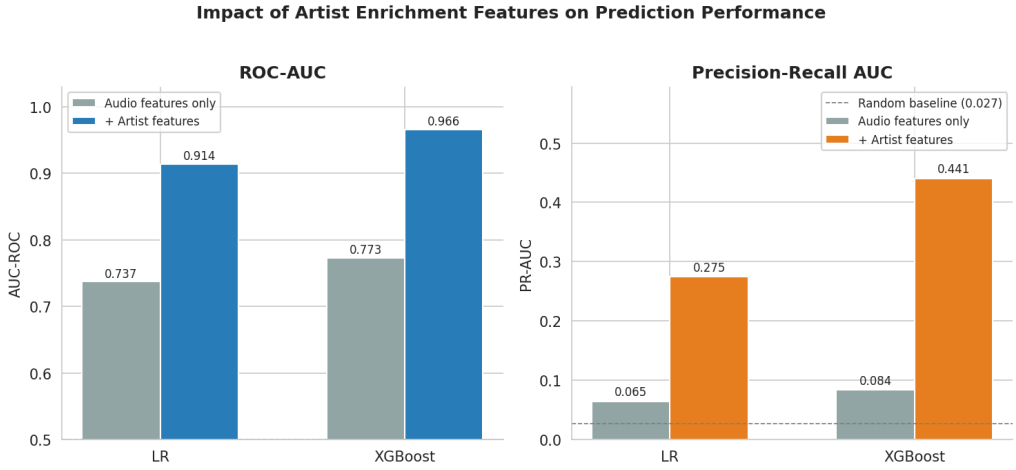


Figure 3: Comparative Performance of Audio-Only vs. Artist-Feature-Enriched Model



3.2 Survival Analysis: Chart Longevity (RQ2)

We modeled longevity – defined as weeks on the Hot 100 – using a Log-OLS regression. By regressing the logarithm of weeks on the chart (*log1p[wks_on_chart])*, we corrected for the heavy right-skew of the raw data, where a small number of "mega-hits" stay on the chart far longer than the average song.

As a robustness check, we compared regression results using a Cox Proportional Hazards (Cox PH) model to account for right-censoring in the chart data. We verified the proportional hazards assumption

using Schoenfeld residuals (see Appendix). Because the "mode" variable (major vs. minor key) violated this assumption, we utilized a stratified model. This approach allowed the baseline exit risk to differ by musical modality without forcing an identical effect over the entire seven-decade dataset. Comparing these two models allows us to distinguish true structural findings from statistical artifacts.

IV. Results

4.1 Research Question 1: Chart Entry Classification

Which audio features and song attributes predict whether a track reaches the Hot 100?

Two machine learning models: logistic regression and XGBoost, were trained on a dataset of 62,712 tracks and validated against a dataset of 15,678 tracks to predict whether a track would enter the Billboard Hot 100. Features spanned both artist profile (popularity scores, listener counts, catalog depth) and audio characteristics (valence, tempo, acousticness, instrumentalness, and others). Models were evaluated on a held-out test set using AUC-ROC, average precision, and a decision threshold analysis that translates model scores into practical record label screening recommendations.

Finding 1: Who releases the song matters more than the song itself. Artist peak popularity and Last.fm listener count are the two strongest predictors of chart entry. An established artist with a large existing audience starts with a significant structural advantage before a single note is evaluated.

Finding 2: Positive mood and the presence of vocals increase charting probability. The Billboard Hot 100 is structurally oriented toward vocal-heavy genres, such as pop and hip-hop, and our model clearly reflects this bias. We found that emotionally positive, vocal-forward tracks are significantly more likely to reach the chart, while a lack of vocal content (instrumentalness ~1.0) serves as a primary deterrent.

Finding 3: Being a US artist independently boosts chart odds. The Billboard Hot 100 has a well-documented bias toward domestic artists and US radio markets, and the model picks this up clearly.

Figure 4 shows the precision-recall curves for both classifiers. The random baseline of 0.027 corresponds to a 2.75% positive rate, and XGBoost achieves a 16-fold lift with a PR-AUC of 0.441 compared to logistic regression at 0.275. Figure 5 shows the ROC curves, plotting the true positive rate against the false positive rate. Both models perform above the 0.500 random baseline, with XGBoost (AUC = 0.966) outperforming logistic regression (AUC = 0.914).

The steep upward trajectory of these curves indicates the models successfully recover a large share of genuine chart hits while maintaining a low rate of false positives. This dynamic makes them useful for A&R teams at record labels, who need to efficiently filter large volumes of music into lists of high-probability hits.

XGBoost outperforms the logistic regression model by 5.2 points in AUC, pointing to meaningful non-linear relationships among the variables. Interactions between specific audio features and artist metrics predict chart entry in complex ways that linear models inherently underweight. Despite XGBoost

being the better operational screening tool, the logistic regression model remains valuable. Its interpretability allows us to understand exactly which individual features increase a track's odds of charting, providing clear insights that a black-box model cannot offer.

Figure 4: Precision-recall curves for both classifiers. The random baseline (gray dashed) lies at PR-AUC = 0.0275, corresponding to the 2.75% positive rate. XGBoost achieves 16-fold lift.

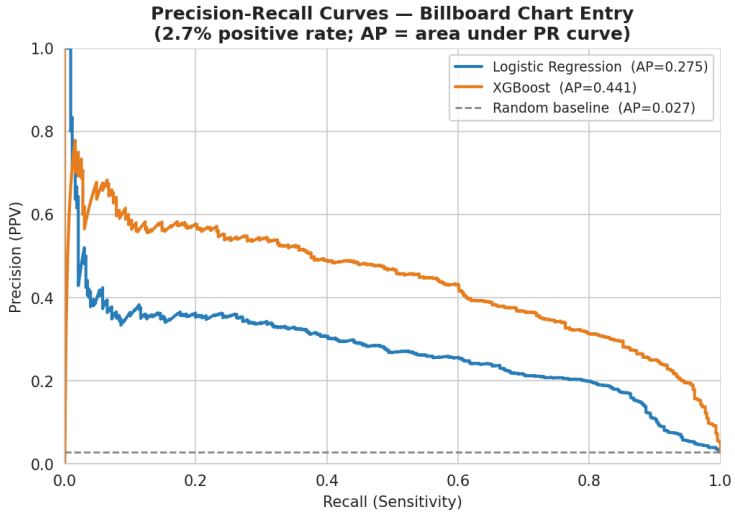


Figure 5: ROC curves for logistic regression (AUC = 0.914) and XGBoost (AUC = 0.966) on the held-out test set.

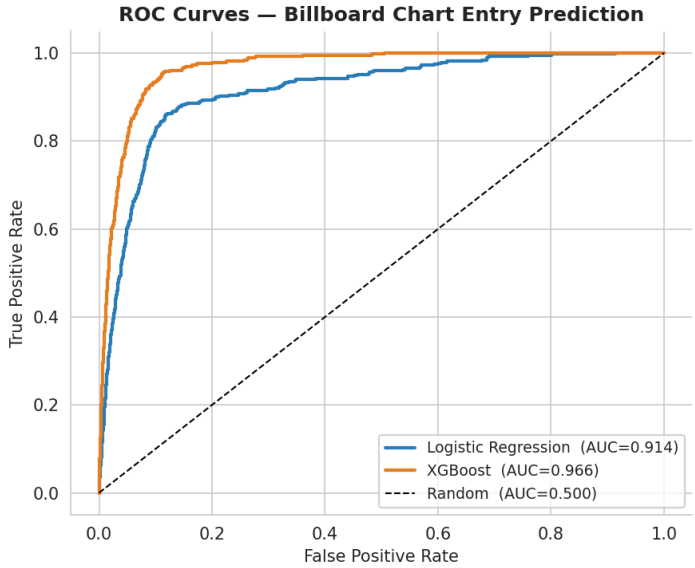
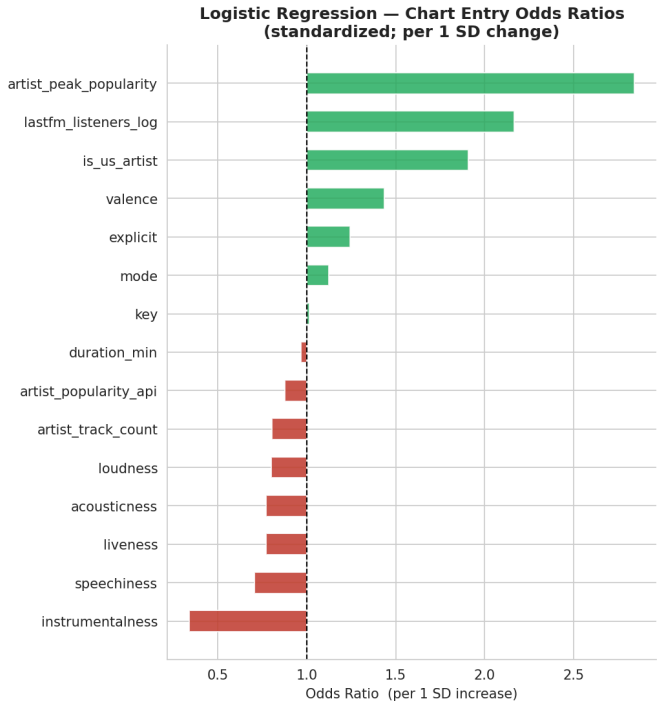


Figure 6 ranks all 15 features by their odds ratio per one standard deviation increase, showing how much each feature shifts the probability of chart entry, and in which direction. Bars to the right of 1.0 increase the odds; bars to the left decrease them.

Figure 6: Audio and artist features on logistic regression model



Note: While artist_peak_popularity is strongly correlated with chart entry, artist_popularity_api (which represents a current point-in-time popularity metric from Spotify at the time of our data collection effort) is negatively correlated with charting given the dataset time period from 1958-2024, where most of the artists charting during this period are no longer popular today.

The most immediate takeaway is the two-tier structure of the chart. The top two features: artist_peak_popularity (OR ≈ 2.8) and lastfm_listeners_log (OR ≈ 2.15), are substantially larger than everything below them, and together they dwarf the entire audio feature set combined. These features are both measures of how commercially established an artist is already.

Is_us_artist (OR ≈ 1.9) comes in third, reflecting the Billboard Hot 100's well-documented structural bias toward domestic artists and US radio markets. Among the audio features, valence (OR ≈ 1.4) is the strongest positive signal, as happier, more emotionally positive tracks are more likely to chart, followed modestly by explicit content and mode.

On the negative side, instrumentalness has the largest downward effect, sitting well below 1.0. Tracks that are primarily instrumental are much less likely to chart on the Hot 100, which is a chart built around vocal pop and hip-hop. Speechiness, liveness, and acousticness all reduce chart odds modestly, while key sits essentially flat at 1.0 – the musical key a song is in carries almost no predictive signal.

The practical message for a label is the same one that runs through this entire analysis: invest in understanding an artist's existing audience before evaluating their sound. The sound matters at the margin, but the audience is what the model is really detecting.

4.2 Research Question 2: Chart Longevity

Among tracks that chart, which features predict how long they remain on the Hot 100?

We modeled log(weeks_on_chart) for Billboard Hot 100 songs using OLS regression, progressively enriching our feature set, then validated findings with a Cox proportional hazards survival framework.

Finding 1: Audio features alone explain very little. Our initial hypothesis was that the "musicality" (audio features) of a song may explain success. Our baseline OLS used Spotify audio features to predict log weeks_on_chart for 2,157 songs. Though jointly significant ($p < 0.001$), the model explained only ~5% of variance ($R^2 = 0.047$). Longer and major-key tracks stayed longer, while higher speechiness, acousticness, and instrumentalness predicted shorter runs. The Cox baseline concordance of 0.551 similarly shows audio features alone barely outperform chance at discriminating short- from long-charting tracks. This finding led us to pursue other features, including genre and artist.

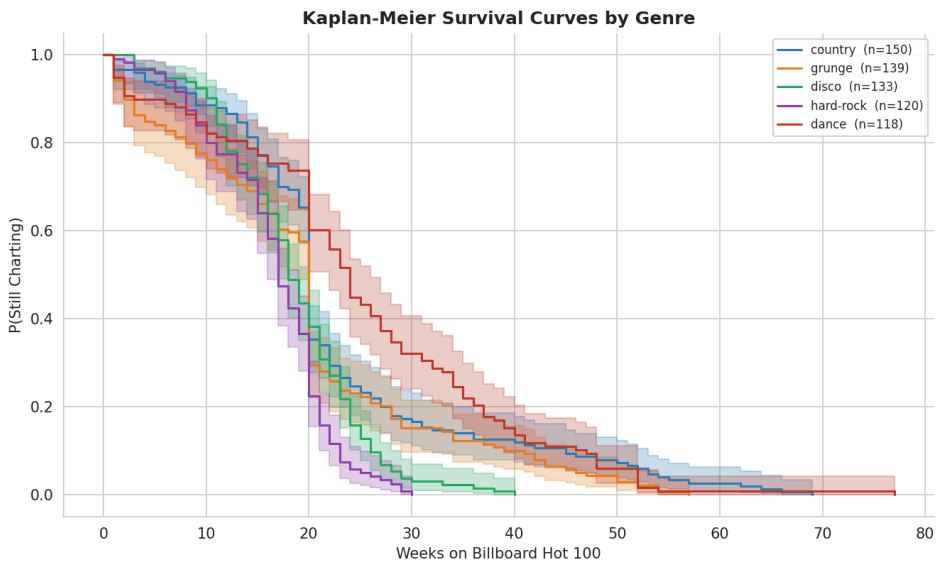
Finding 2: Genre and artist identity are the dominant drivers. Adding genre dummies raised R^2 from 0.05 to 0.27, indicating genre captures audience segments, radio formats, and playlist dynamics that audio alone cannot. Adding artist-level controls (followers, monthly listeners, Last.fm scrobbles) yielded $R^2 \approx 0.15$ without genre and $R^2 \approx 0.23$ with genre included. In our richest specification, which incorporated sentiment, lyric features, and additional artist metadata, R^2 reached 0.24, though sentiment and lyric variables contributed almost no incremental signal in the linear framework.

Finding 3: Cox survival model confirms and extends these findings.

The Cox proportional hazards model (concordance = 0.753 on 856 charted tracks) substantially outperforms the audio-only baseline and confirms that artist identity matters for longevity, not just chart entry. The strongest predictor is decade_idx (decade of chart entry; HR = 0.514, $p < 0.001$): tracks from more recent decades tend to remain on the chart longer on a per-week basis, likely reflecting the streaming era's tendency to produce outlier mega-hits with unprecedented chart runs.

As illustrated by Figure 8, the probability of remaining on the chart declines sharply in the first 10 weeks, with genre-level differences reflecting format and rotational differences across radio markets.

Figure 8: Kaplan-Meier Survival Curves by Genre



Among artist features, higher historical peak popularity predicts longer chart stays (HR = 0.835, $p < 0.001$), while larger catalog size is associated with faster exit (HR = 1.115, $p = 0.005$). This suggests that while a proven track record of commercial success helps sustain chart presence, artists with larger catalogs may see attention spread more thinly across releases, leading to faster turnover for any individual song. Loudness is the only significant audio feature (HR = 0.900, $p = 0.012$): louder tracks stay on the chart longer.

Finding 4: Models do not generalize forward in time. We evaluated all OLS specifications using temporal train-test splits (cutoffs at 2015 and 2019). Every model produced negative out-of-time R^2 – performing worse than predicting the mean for all future songs. The richest in-sample model ($R^2 = 0.48$) achieved a test R^2 of -0.32. This is consistent with the Cox model's decade-index finding: the music industry's structural evolutions (changing streaming platforms, playlist algorithms, and consumption patterns) means historical feature-outcome relationships do not extrapolate reliably to future releases.

These models tell us that artist reach and genre dominate chart longevity, and streaming has fundamentally accelerated chart turnover. But they should not be used as forecasting tools for new releases.

V. Conclusions

Our song charting analysis revealed several key insights regarding the likelihood of a song reaching the Billboard Hot 100 and its expected longevity on the charts:

1. Audio features are minimally predictive of songs' odds of charting on the Billboard Hot 100
2. Primarily artist popularity, and secondarily genre, most influence Billboard Hot 100 charting
3. Billboard Hot 100 charting is biased toward domestic U.S. artists
4. All factors influencing charting longevity have seen absolute correlation decrease over time

To better understand the likelihood of commercial success, record labels should understand an artist's existing audience before evaluating their sound. Audio features barely outperform random at a 2.75% positive rate when stripped of artist context. Of the audio signals that do matter, avoiding highly instrumental production and producing emotionally upbeat, vocal-forward material would increase probability of chart success, but these are second-order effects compared to the artist's existing popularity.

Furthermore, streaming has compressed chart lifespans by roughly half since the 1980s. Genre and artist identity dominate how long a song stays on the chart, but streaming has dramatically accelerated chart turnover, with each additional decade corresponding to a 49% increase in per-week exit hazard. The Cox model confirms decade index as the only predictor consistent across all specifications., and while genre and artist identity explain meaningful variance in-sample, no model generalizes to future releases – suggesting the factors driving chart longevity are shifting faster than historical data can capture.

The takeaway for a label is that high volume output of songs by popular artists is more likely to net greater overall time on the Billboard Hot 100 than a smaller number of 'higher-quality' songs. The structural evolution of music listening to a streaming-first environment has effectively reduced societal attention spans, and stronger commercial partnerships and better marketing are more important than ever.

Our findings are bounded by the inability to observe real-time streaming and playlist dynamics that may better predict song success (what else listeners are listening to at a given time, what other songs are charting in previous weeks, etc.) and our data limitation to US charts. Two directions would meaningfully extend this work. First, incorporating real-time playlist placement and streaming velocity data would directly address the failure to predict future releases. Second, extending the framework to international charts such as the UK Singles Chart would test whether the US artist origin effect reflects genuine market structure or is an artifact of building models on a US-domestic chart.

We grant permission for this project to be kept and used as an example for future classes.

VI. Appendix

References

The following sources were used in this analysis. All Kaggle datasets are public and available without registration for download.

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[8] Pachet, F., & Roy, P. (2008). Hit Song Science Is Not Yet a Science. Proceedings of the International Society for Music Information Retrieval Conference (ISMIR 2008).

Data cleaning pipeline

Data cleaning step	**Number of tracks**	**Questions Addressed Using this Dataset**
Base dataset with audio features	114,000 tracks from Spotify dataset	-
Removing duplicates	78,390 unique tracks	Q1
Adding artist features	61,871 unique tracks with artist features	Q1
Joining with Billboard Hot 100	2,157 charted tracks	Q1, Q2
Adding artist features	1,251 charted tracks with artist features	Q2
Added sentiment/lyric features	856 charted tracks with artist + sentiment/lyric features	Q2

Variable Definitions

Table X.X: Model Feature Definitions

All features listed below are those used in at least one model in the final v6 pipeline. Audio features are sourced from the Spotify Tracks Dataset; artist-level features from the Spotify Web Developer API and the Music Artists Popularity dataset; temporal and sentiment features are derived variables.

Feature	Type / Range	Model(s) Used	Definition
valence	[0, 1]	All	Spotify's measure of musical positiveness. High values correspond to happy, euphoric, or uplifting tracks; low values to sad, angry, or tense tracks.
acousticness	[0, 1]	All	Confidence score (0–1) that the track was recorded acoustically rather than electronically produced. Derived from Spotify's audio analysis.
loudness	dB (~−60 to 0)	All	Overall average loudness of the track in decibels (LUFS). Values closer to 0 indicate louder tracks. Negative by convention.
speechiness	[0, 1]	All	Presence of spoken words. Values above 0.66 indicate speech-dominated content; below 0.33 indicate music with no speech; 0.33–0.66 indicate mixed content (e.g., rap).
instrumentalness	[0, 1]	All	Probability that the track contains no vocal content. Values above 0.5 are classified as instrumental. The dominant audio barrier to chart entry (OR = 0.34, SHAP = 0.629).
liveness	[0, 1]	All	Probability that the track was recorded in front of a live audience. Higher values suggest a live performance recording.
mode	0 or 1	All (stratum in Cox)	Musical modality. 1 = major key; 0 = minor key. Used as a stratification variable in Cox PH because it fails the Schoenfeld proportional hazards test.
key	0–11	All	Pitch class of the track using standard Pitch Class notation (0 = C, 1 = C#/D ♭ , …, 11 = B). Treated as a linear numeric feature; circular embedding was not applied.
explicit	0 or 1	All	Whether the track has an explicit content flag on Spotify. 1 = explicit; 0 = clean. Explicit tracks are 24% more likely to chart per the logistic regression odds ratio.
duration_min	minutes (capped 10)	All	Track duration in minutes, derived from duration_ms / 60000. Capped at 10 minutes to exclude podcast episodes and audiobook entries present in the Kaggle source.
artist_peak_popularity	0–100	All	Historical peak Spotify popularity score for the track's primary artist, from the Spotify API. The strongest predictor in all models (SHAP = 1.618, OR = 2.84 per 1 SD).
artist_popularity_api	0–100	All	Current Spotify popularity score for the artist at the time of data collection. Highly collinear with artist_peak_popularity (VIF = 14.1); individual coefficient signs should not be interpreted in isolation.
artist_track_count	integer	All	Number of tracks on Spotify attributed to the primary artist. Proxy for catalog size and career stage.
lastfm_listeners_log	log1p scale	All	log1p-transformed cross-platform monthly listener count from the Last.fm / MusicBrainz dataset (1.47M artists). Covers 65.8% of artists by name match. The second-strongest predictor (SHAP = 1.153, OR = 2.17 per 1 SD).
is_us_artist	0 or 1	All	Binary indicator for US-based artist, derived from the MusicBrainz country field. Artists with unknown nationality are coded 0 (conservative imputation). US artists are 91% more likely to chart (OR = 1.91).
decade_idx	0–7	Cox PH, OLS only	Ordinal decade index derived from each charted track's first chart appearance: floor((chart_entry_year − 1950) / 10). 0 = 1950s, 1 = 1960s, …, 7 = 2020s. Undefined (NaN) for

Feature	Type / Range	Model(s) Used	Definition
			non-charted tracks. The strongest longevity predictor (HR = 0.514, $p < 0.001$).
sentiment_compound	[−1, +1]	Cox PH, OLS only	VADER aggregate sentiment score for the track's Billboard lyrics. Computed as the normalized, weighted sum of positive, negative, and neutral lexicon scores. Coverage: 40.1% of charted tracks (n = 856).
sentiment_pos	[0, 1]	Cox PH, OLS only	Fraction of VADER-identified positive words in the track's lyrics. Not statistically significant in v6 longevity models ($p = 0.366$). Fails Schoenfeld PH test ($p = 0.393$ rank test).
sentiment_neg	[0, 1]	Cox PH, OLS only	Fraction of VADER-identified negative words in the track's lyrics. HR = 1.088, $p = 0.051$ (marginal); directionally consistent with shorter chart runs for tracks with more negative lexical content.
lyric_word_count	integer	Cox PH, OLS only	Total word count of the track's Billboard lyrics. Not statistically significant in v6 longevity models ($p = 0.592$).

Features removed for multicollinearity (VIF > 10): energy (VIF = 15.1), danceability (VIF = 12.4), tempo (VIF = 10.7). Features in dataset but excluded for insufficient coverage: is_male_artist, artist_age, is_mainstream_genre (<58% of charted tracks; <1% of full universe). Full VIF table in Appendix A.

VIF Table

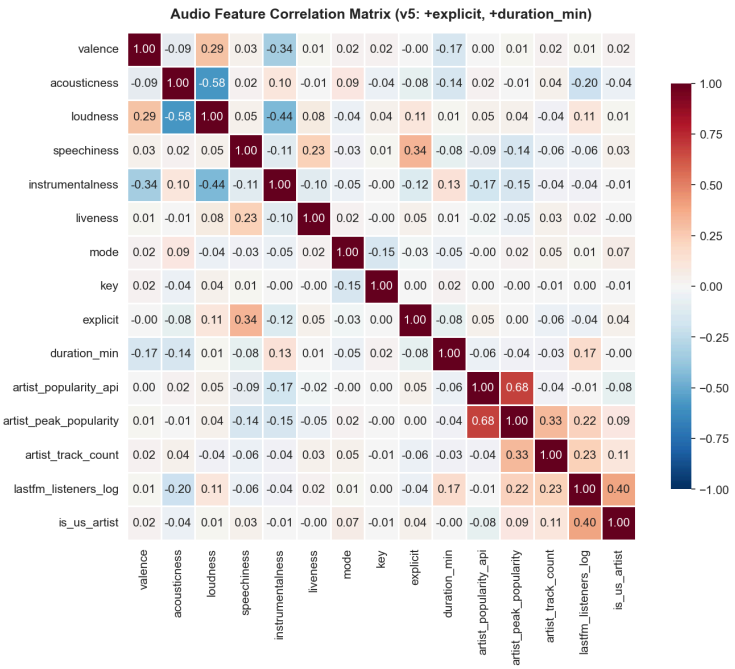
Table X.X: Feature VIF

Feature	VIF
artist_peak_popularity	**21.25**
artist_popularity_api	**14.12**
loudness	6.72
duration_min	6.28
lastfm_listeners_log	5.18
valence	3.75
acousticness	3.17
key	3.04
mode	2.65
liveness	2.31
speechiness	1.89
instrumentalness	1.87
artist_track_count	1.77
is_us_artist	1.40
explicit	1.27

Features with VIF > 10 flagged in red. energy (VIF = 15.1), danceability (VIF = 12.4), and tempo (VIF = 10.7) were removed from the final model.

Supporting Exhibits

Figure 1: Audio Feature Correlation Matrix



Research Question 1 - Additional Supporting Analysis

This table focuses specifically on XGBoost model. Each row represents a different threshold setting, essentially how confident the model needs to be before it flags a track, and shows what a record label team would experience at that setting in terms of workload, accuracy, and coverage.

As the threshold rises from 0.05 to 0.50, the number of flagged tracks shrinks from 4,044 to 1,684, meaning the team reviews fewer songs. Precision rises from 10.5% to 23.5%, meaning a higher share of what they do review turns out to be a genuine hit. Recall, however, falls from 98.1% to 91.9%, meaning they start missing some real hits that didn't make the shortlist.

Observe how slowly recall falls as the threshold tightens. Going from 0.05 all the way to 0.30 cuts the review workload by nearly half, from 4,044 tracks down to 2,143, while recall only drops from 98.1% to 95.6%. The label is giving up less than 3% of hit coverage in exchange for reviewing almost half as many tracks, which is a favourable tradeoff.

Threshold of 0.20 is recommended as it sits in the sweet spot: 2,506 tracks flagged (16% of the test catalog), 96.1% recall, and 16.5% precision. Whether that tradeoff is acceptable depends entirely on how large and resourced the record label team is, and how aggressively the label is competing for new signings.

Table X.X: Model Performance: Audio-Only Baseline vs. Full Model (Audio + Artist Features)

Model	Task	Metric	Audio Only	Full Model	PR-AUC (Full)	CV / Notes
Logistic Regression	Chart Entry	AUC-ROC	0.737	**0.914**	0.275	0.914 ± 0.004 (5-fold CV)
XGBoost	Chart Entry	AUC-ROC	0.773	**0.966**	0.441	Early stop @ iter 189
Cox PH	Longevity	C-statistic	0.551	**0.753**	—	Mode stratified; n = 856
Log-OLS (robustness)	Longevity	R^2	0.044	**0.212**	—	log1p(wks); n = 856

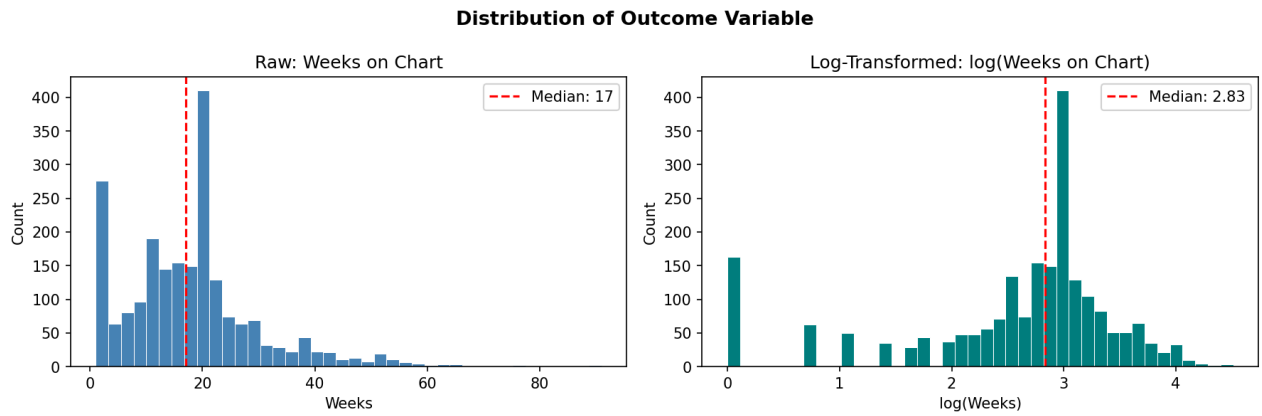
Table X.X: Logistic Regression Odds Ratios (All Features)

Feature	Coefficient	Odds Ratio
artist_peak_popularity	+1.045	2.84
lastfm_listeners_log	+0.773	2.17
is_us_artist	+0.647	1.91
valence	+0.362	1.44
explicit	+0.217	1.24
mode	+0.117	1.12
key	+0.013	1.01
duration_min	-0.034	0.97
artist_popularity_api	-0.132	0.88 *
artist_track_count	-0.217	0.80
loudness	-0.221	0.80
acousticness	-0.260	0.77
liveness	-0.262	0.77
speechiness	-0.347	0.71
instrumentalness	-1.081	0.34

** artist_popularity_api sign reversal is a collinearity artifact due to co-occurrence with artist_peak_popularity (VIF = 14.12). Only the combined directional signal (established artists more likely to chart) is interpretable.*

Research Question 2 - Additional Supporting Analysis

Figure X.X: Distribution of Chart Longevity



We see in the data that the number of weeks that a given song spends on the Billboard Hot 100 chart is right skewed. Given the skewness of the raw outcome, we log-transformed the dependent variable to satisfy OLS normality assumptions.

Figure X.X: Cox PH – Hazard Ratios

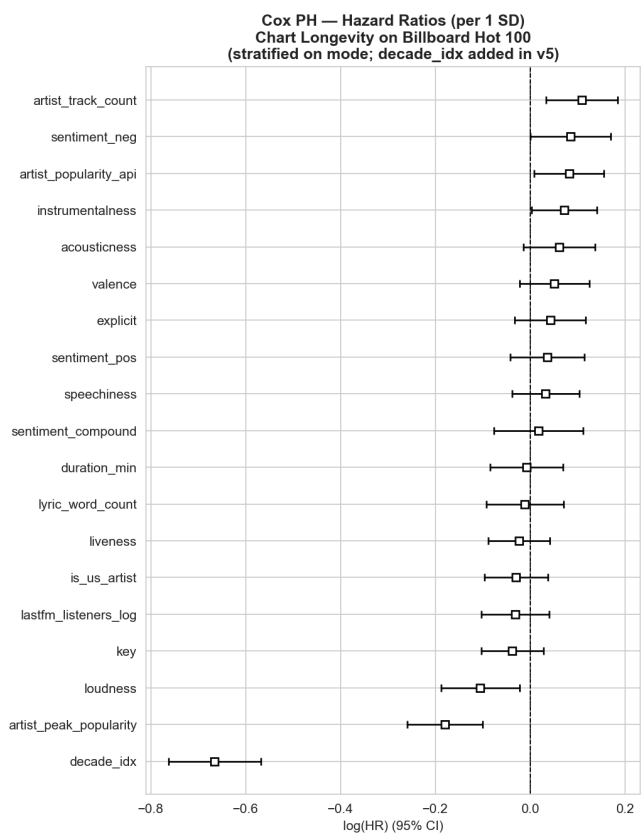


Table X.X: Cox PH Significant Coefficients

Feature	Coef	HR	p-value	Interpretation
decade_idx	-0.666	**0.514**	<0.001	49% lower exit hazard per decade
artist_peak_popularity	-0.180	**0.835**	<0.001	Higher peak = longer runs
artist_track_count	+0.109	**1.115**	0.005	Larger catalog = shorter runs
artist_popularity_api	+0.082	**1.085**	0.030	Current popularity does not sustain chart
loudness	-0.106	**0.900**	0.012	Louder = longer chart runs
instrumentalness	+0.071	**1.074**	0.043	Instrumental = exits faster

HR = hazard ratio. HR > 1 indicates faster exit from chart (shorter run); HR < 1 indicates slower exit (longer run). Mode handled by stratification. Note that artist_peak_popularity and artist_popularity_api have opposing effects. These variables are highly collinear (VIF > 14), so individual coefficients should be interpreted cautiously. However, one hypothesis is that historical peak popularity captures durable fanbase depth, which sustains chart presence, while current API popularity, conditional on peak, may partially reflect short-lived viral attention that fades faster.

We ran multiple regressions based on our hypotheses.

<u>Model 1: Audio features only</u>

Figure X: Model output for regression of chart longevity on audio features

```
Outcome: log(weeks_on_chart)
N = 2,157
R2 = 0.0471
Adj. R2 = 0.0409
F-stat p-value = 9.0608e-16

        variable      coef      std_err         t       pvalue sig
           const  1.911614 3.469475e-01  5.509807 4.024046e-08 ***
    danceability  0.383871 1.793722e-01  2.140083 3.246071e-02   *
          energy  0.084594 1.968493e-01  0.429741 6.674273e-01
             key -0.013001 5.896987e-03 -2.204654 2.758437e-02   *
        loudness -0.008860 9.833925e-03 -0.900929 3.677274e-01
            mode  0.094108 4.727710e-02  1.990558 4.665634e-02   *
     speechiness -1.184618 3.474077e-01 -3.409879 6.619599e-04 ***
    acousticness -0.370658 1.096781e-01 -3.379506 7.391748e-04 ***
instrumentalness -0.512753 2.166894e-01 -2.366303 1.805529e-02   *
        liveness -0.322821 1.547884e-01 -2.085566 3.713605e-02   *
         valence  0.225307 1.147312e-01  1.963778 4.968501e-02   *
           tempo -0.001648 7.776561e-04 -2.119639 3.415120e-02   *
  time_signature  0.080452 6.573796e-02  1.223822 2.211540e-01
     duration_ms  0.000001 3.812570e-07  3.406642 6.698171e-04 ***
        explicit -0.095054 7.568908e-02 -1.255844 2.093094e-01
```

Audio features alone explain only about 5% of the variation in log(weeks on chart), with danceability, speechiness, acousticness, and duration among the significant predictors.

<u>Model 2: Genre + audio features</u>

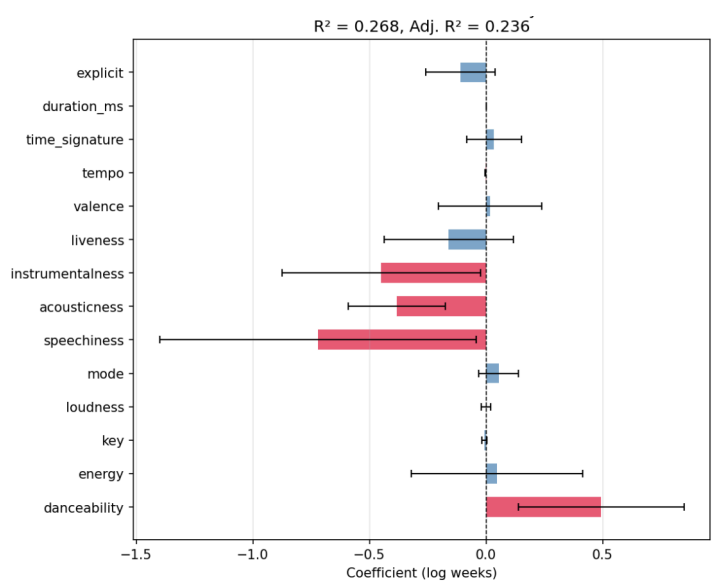
Figure X: Model output for regression of chart longevity on audio features + genre

```
Outcome: log(weeks_on_chart)
N = 2,157
R2 = 0.2679
Adj. R2 = 0.2364
F-stat p-value = 1.2585e-87

               variable          coef      std_err         t       pvalue sig
                  const  2.256758e+00 3.611434e-01  6.248924 5.004677e-10 ***
           danceability  4.936682e-01 1.814187e-01  2.721154 6.559990e-03  **
                 energy  4.659623e-02 1.878051e-01  0.248110 8.040743e-01
                    key -7.757602e-03 5.355952e-03 -1.448408 1.476547e-01
               loudness  1.521761e-04 9.946156e-03  0.015300 9.877943e-01
                   mode  5.475919e-02 4.344526e-02  1.260418 2.076609e-01
            speechiness -7.205122e-01 3.463285e-01 -2.080430 3.760904e-02   *
           acousticness -3.829846e-01 1.064640e-01 -3.597314 3.290724e-04 ***
       instrumentalness -4.487804e-01 2.170904e-01 -2.067252 3.883465e-02   *
               liveness -1.595880e-01 1.406946e-01 -1.134286 2.568060e-01
                valence  1.695522e-02 1.132718e-01  0.149686 8.810268e-01
                  tempo -1.654138e-03 7.147603e-04 -2.314256 2.075130e-02   *
         time_signature  3.488614e-02 5.970526e-02  0.584306 5.590783e-01
            duration_ms  5.795095e-07 3.781223e-07  1.532598 1.255279e-01
               explicit -1.093070e-01 7.567276e-02 -1.444470 1.487583e-01
```

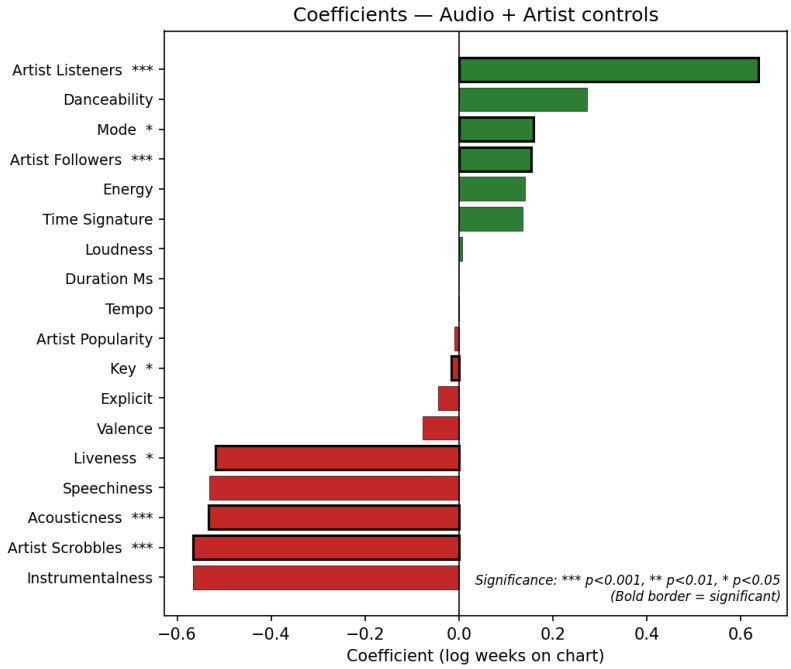
Note: 75 genres were left out of the regression output for space reasons.

Figure X: Effect of audio features on chart longevity (error bars = 95% CI; red = significant at $p < 0.05$)



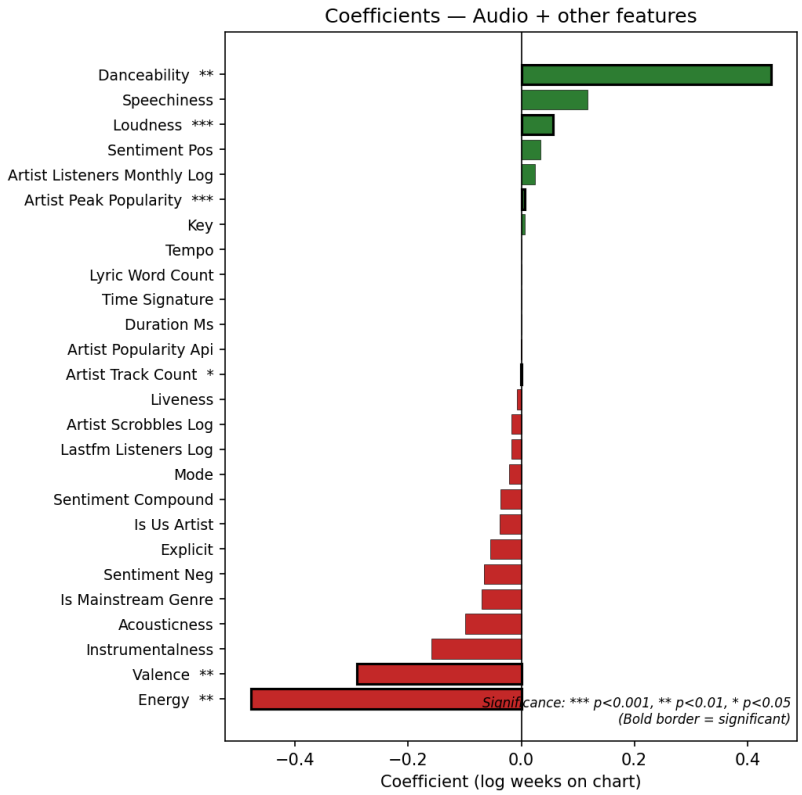
Adding genre dummies raises R^2 to about 27%, with genres like country and dance associated with longer chart life and club with shorter.

Model 3: Audio features + genre + artist controls



Including both artist controls and genre dummies pushes R^2 to about 23%, with artist and genre effects both significant.

Model 4: Audio features + genre + artist controls + sentiment/lyrics



Using the full feature set (artist, metadata, sentiment, lyrics) on the subset with complete data yields R^2 of about 24%, with artist peak popularity and several audio features (e.g., loudness, valence) significant. Adding genre does not improve fit beyond the audio + other model (R^2 unchanged at about 24%).

<u>Temporal Validation Model</u>

Year	Sample	Model	R-squared (Train)	R-squared (Test)
2015	Train N=1633, Test N=524	Model 1 (Audio only)	0.055	-0.057
2015	Train N=367, Test N=145	Model 2 (Audio + other)	0.363	-0.404
2015	Train N=367, Test N=145	Model 3 (Audio + other + genre)	0.479	-0.323
2019	Train N=1867, Test N=290	Model 1 (Audio only)	0.066	-0.119
2019	Train N=446, Test N=66	Model 2 (Audio + other)	0.324	-0.103
2019	Train N=446, Test N=66	Model 3 (Audio + other + genre)	0.452	-0.105

For all three model types (audio only, audio + other, audio + other + genre), out-of-time R^2 on 2015+ data is negative, so predictions are worse than using the test-period mean. The same pattern holds for 2019, indicating that this is not a matter of recency bias.